Polynomial Regression: Model Selection and Evaluation

Machine Learning Assignment | **BT2024134**

1 Approach

I trained regularized polynomial regression models independently for the two datasets: six inputs for `var1` and three for `var2`. Each dataset has 1,000 labeled training rows. I checked feature columns and non-finite values, retained repeated observations, and used test data only to export predictions.

Splitting and evaluation. I grouped identical input vectors before splitting, using `GroupShuffleSplit` (20% holdout, seed 134). Model selection used five-fold `GroupKFold` on development rows only. I ranked candidates by pooled out-of-fold (OOF) MSE, with OOF R^2 as an additional metric. The holdout was kept untouched until model selection was complete.

Table 1: Saved data partitions.

Problem	Development	Holdout	Validation fold sizes
`var1`	800	200	160, 160, 160, 160, 160
`var2`	796	204	160, 159, 159, 159, 159

Preprocessing per fit. Standardize original inputs → polynomial expansion without a bias column → standardize expanded columns and target → fit → transform predictions back to the original target scale. All scaling parameters were learned from the corresponding training fold only.

Candidates. I compared OLS, Ridge, Lasso, and Elastic Net across polynomial degrees, regularization strengths (α), and Elastic Net L1 ratios (ρ). OLS and Ridge used SVD; Lasso and Elastic Net used iterative optimization. Candidates required five finite, converged folds. The broad search used solver tolerance 10^{-5}; the narrow search and final fits used 10^{-7}.

2 Broad search

I tested degrees 4–6 for `var1` and 10–13 for `var2`. The mixed/Lasso alpha ranges were 0.0003–0.1 and 0.0001–0.01 respectively; Ridge used nine log-spaced alphas from 10^{-6} to 10^{2}. In total: **706 configurations / 3,530 fold fits**; 672 configurations were eligible, while 34 had at least one unconverged fold (82 unconverged fits total).

Table 2: Best eligible broad-search candidate at each degree.

Problem	Degree	α	ρ	Pooled OOF MSE
`var1`	4	0.0035	0.95	0.701610
	5	0.0030	1.00	**0.371650**
	6	0.0035	1.00	0.387813
`var2`	10	0.0004	0.90	0.375608
	11	0.0002	1.00	**0.330569**
	12	0.0006	0.90	0.352191
	13	0.0003	1.00	0.334807

Search decisions. For `var1`, degree 5 beat degree 6 on every broad-search fold, so I focused the narrow search on degree 5 while keeping degree 6 as a challenger. For `var2`, degrees 11 and 13 were close, and a degree-11 mixed Elastic Net candidate was nearly tied with Lasso; therefore I retained both near-Lasso and mixed-penalty regions, plus smaller degree-12/13 checks.

3 Narrow search and selection

I refined promising (α, ρ) regions, carried forward broad winners and model-family controls, and revisited three previously unconverged `var2` configurations. After deduplication, the narrow searches evaluated **93 configurations for `var1`** and **192 for `var2`** (1,425 fold fits); all recorded narrow fits converged.

Table 3: Narrow-search regions.

Problem	Degree	α range	ρ range	Candidates
`var1`	5	0.00225–0.004	0.75–1.00	54
	6	0.0025–0.0045	0.85–1.00	35
`var2`	11	0.000125–0.0003	0.80–1.00	35
	11	0.00025–0.0006	0.45–0.75	56
	12	0.0004–0.0008	0.75–1.00	25
	13	0.0002–0.0005	0.80–1.00	35
	13	0.00035–0.0008	0.50–0.75	36

Selected models. `var1` used degree-5 Lasso ($\alpha = 0.003125$). `var2` used degree-11 Elastic Net ($\alpha = 0.0002$, $\rho = 0.9$): narrow-search OOF MSE was 0.330639 versus approximately 0.330672 for degree-11 Lasso at the same alpha. That margin is small; selection followed the lowest eligible narrow-search MSE rather than assuming a meaningful gap. The lower broad-search Lasso error used the looser solver tolerance and was not used for final ranking.

4 Final evaluation and export

After freezing hyperparameters, I trained on development rows and evaluated the reserved holdout. I then refitted each chosen configuration on all 1,000 labeled rows to produce test predictions.

Table 4: Final recorded settings and metrics.

Setting or metric	`var1`	`var2`
Model	Lasso	Elastic Net
Degree	**5**	**11**
Alpha	0.003125	0.000200
L1 ratio	1.00	0.90
Expanded features	461	363
Development OOF MSE	0.371575710	0.330639114
Development OOF R^2	0.962565266	0.992420615
Holdout MSE	0.332857104	0.253029670
Holdout R^2	0.968185636	0.994930274

Each output contains 1,000 predictions in a single `y` column in original test-row order: `BT2024134_pred_var1.csv` and `BT2024134_pred_var2.csv`. Holdout metrics evaluate the development-trained models, not the final full-data fits.